

Exercise 2

AIM :

To create a C program for the below requirements

1. Create two source file main.c & display.c
2. Create a header file main.h
3. In the header file create a macro "STAGE", also by using conditional compilation for "STAGE" value as 1 & 2 define a macro for "AREA". if STAGE = 1, the AREA should have the expression for area of square, if STAGE = 2, the AREA should have the expression of area of circle.
4. In main.c create a global variable for the radius(for circle) and side (square) by using another conditional compilation
5. in display.c print the output of area in this file based on the AREA macro. the global variables should accessed as extern variable

ALGORITHM :

Step 1: START

Step 2: Create header file main.h with include guard

Step 3: Define macro STAGE with value 1 or 2

Step 4: Implement conditional compilation:

IF STAGE == 1 THEN

Define AREA(s) as $(s * s)$

Define SHAPE_NAME as "Square"

ELSE IF STAGE == 2 THEN

Define PI as 3.14159

Define AREA(r) as $(PI * r * r)$

Define SHAPE_NAME as "Circle"

ELSE

Generate compilation error

Step 5: Create main.c file:

IF STAGE == 1 THEN

Declare global variable 'side' as integer

ELSE IF STAGE == 2 THEN

 Declare global variable 'radius' as float

ENDIF

Call display_area() function

Step 6: Create display.c file:

Declare external variables using extern keyword

IF STAGE == 1 THEN

 Access 'side' variable

 Calculate area using AREA(side) macro

 Display side and area

ELSE IF STAGE == 2 THEN

 Access 'radius' variable

 Calculate area using AREA(radius) macro

 Display radius and area

ENDIF

Step 7: Compile all files together

Step 8: Execute the program

Step 9: STOP

OUTPUT :

```
D:\C programming\DHAVAMANI_EX2>gcc main.c display.c -o shape_area.exe
D:\C programming\DHAVAMANI_EX2>shape_area.exe

Shape Area Calculator
Current STAGE: 1 (Square)

Calculating area for Square:
Side length: 5 units
Area = 25 square units

D:\C programming\DHAVAMANI_EX2>gcc main.c display.c -o shape_area.exe
D:\C programming\DHAVAMANI_EX2>shape_area.exe

Shape Area Calculator
Current STAGE: 2 (Circle)

Calculating area for Circle:
Radius: 5.00 units
Area = 78.54 square units
```